

Selecting Random Samples

Pulling names/numbers/etc out of a hat

- This is a classic description of an SRS that can be used on the AP exam to describe selecting a random sample, but it is rarely done in practice due to it being so time consuming.
- Write down all the names or numbers each on their own slip of paper. Then put all the pieces of paper into a hat and pull out n slips (mention with or without replacement).

Calculator

- MATH $\rightarrow$ PRB $\rightarrow$ 5:randInt(lower, upper, n)
- Can be used to describe on the AP Exam but not really used for “doing” random samples on the AP Exam because there is no way to “check” for it.
- In practice, computer generated numbers are the least time consuming.

Table of Random Digits

- This used to be called “Table D” on the AP exam, but has since been removed. It is now given as a long string of digits 0-9. The digits are grouped in 5’s to make it easier to read, but it has no significant meaning.

Choosing SRS with table:

1. Label: Assign a number label to every individual in the population.
2. Random Digits: Start at the very first number and identify how many digits you will take at a time.
3. Stop: Indicate when you should stop sampling (toss out repeated numbers, or numbers out of your range)
4. Identify sample: use the random #'s to identify subjects to be selected from your population. This is your sample!

Example 1) The school newspaper is planning an article on family-friendly places to stay over spring break at nearby beach town. The editors intend to call 4 randomly chosen hotels to ask about their amenities for families with children. They have an alphabetized list of all 28 hotels in the town. ~~Starting at line 140~~, find an SRS of 4 hotels. Describe how you would select your SRS and then collect your sample.

01 Aloha Kai	08 Captiva	15 Palm Tree	22 Sea Shell
02 Anchor Down	09 Casa del Mar	16 Radisson	23 Silver Beach
03 Banana Bay	10 Coconuts	17 Ramada	24 Sunset Beach
04 Banyan Tree	11 Diplomat	18 Sandpiper	25 Tradewinds
05 Beach Castle	12 Holiday Inn	19 Sea Castle	26 Tropical Breeze
06 Best Western	13 Lime Tree	20 Sea Club	27 Tropical Shores
07 Cabana	14 Outrigger	21 Sea Grape	28 Veranda

~~12975~~ ~~13258~~ ~~13048~~ ~~45144~~ ~~72321~~ 81940 00360

12 HOLIDAY INN
 13 LIMETREE

04 BANYAN TREE
 18 SANDPIPER

Introduction to Experiments

Observational Study	Experiment
OBSERVES INDIVIDUALS AND MEASURES VARIABLES BUT <u>DOES NOT</u> ATTEMPT TO INFLUENCE THE RESPONSE	DELIBERATELY IMPOSES A TREATMENT ON INDIVIDUALS TO MEASURE THEIR RESPONSE

- *Experimental Units* – THE INDIVIDUALS ON WHICH THE EXPERIMENT IS DONE
- *Subjects* – WHEN THE EXPERIMENTAL UNITS ARE PEOPLE
- *Treatment* – A SPECIFIC EXPERIMENTAL CONDITION APPLIED TO THE UNITS.

Examples 1-4: Are the following an experiment or an observational study? Explain your answer.

1) A medical team examines the records of 5 large hospitals and compares the survival times of those cancer patients who had surgery versus those who had chemotherapy.

OBSERVATIONAL STUDY

(NO TREATMENT IMPOSED)

2) In a gym class, the effect of exercise on blood pressure is studied by requiring that half of the students walk a mile each day while the other students run a mile each day.

EXPERIMENT (RUN ~~AND~~ ^{VS.} WALK ONE MILE =
IMPOSED TREATMENT)

3) The relationship between weights of bears and their lengths is studied by measuring bears that have been anesthetized.

OBSERVATIONAL STUDY

4) People who smoke are asked to halve the number of cigarettes consumed each day so that any effect on pulse rate can be measured.

EXPERIMENT (ALBEIT A SHITTY ONE)

Example 5 - 9: Determine if it is an observational study or an experiment, and then identify the explanatory and response variables in each situation. *Remember: A **response variable** measures an outcome of a study. An **explanatory variable** attempts to explain the observed outcomes.

5) One effect of alcohol is a drop in body temperature. To study this effect, researchers give several amounts of alcohol to mice, and then measured the change in each mouse's body temperature.

EXPERIMENT:

EXPLANATORY → AMOUNTS OF ALCOHOL

RESPONSE → BODY TEMPERATURE

6) A study is done to try and find the correlation between verbal and math SAT scores. The scientist wants to use the verbal score to predict the math score.

OBSERVATIONAL:

EXPLANATORY → VERBAL

RESPONSE → MATH

7) Some breast cancer patients were given each a new treatment. The patients were closely followed to see how long they lived following surgery.

EXPERIMENT

EXPLANATORY → NEW TREATMENT

RESPONSE → TERM OF SURVIVAL

8) To find out how well a child's height predicts their age a study was done where they measured the heights of a group of children at age 6, wait until they are 16 and then measure their heights again.

OBSERVATIONAL

EXPLANATORY → HEIGHT

RESPONSE → AGE

Question: What is the number one advantage of an experiment over an observational study?

Only with a well-designed, controlled experiment can we say that one variable *causes* a change in another variable. We CANNOT infer causation with an observational study.